

## **Project Documentation:**

Optimizing User, Group, and Role Management with Access Control and Workflows

### **Problem statement:**

In a small project management team consisting of a Project Manager (Alice) and a Team Member (Bob), there is a need to efficiently manage project tasks and ensure accountability throughout the project lifecycle. The current system lacks clear role definitions, access controls, and a structured workflow, leading to confusion regarding task assignments and progress tracking.

### **Objectives:**

- Implement centralized user, group, and role management.
- Define and enforce role-based access control (RBAC).
- Automate workflows for user onboarding, role changes, and access requests.
- Ensure secure, auditable access management aligned with compliance standards.
- Reduce administrative overhead and minimize access-related errors.

### **Core Features:**

#### User Management

- Create, edit, deactivate, and delete user accounts.
- Assign users to one or more groups.
- Associate users with roles dynamically based on department or function.

#### Group Management

- Logical grouping of users (e.g., HR, IT, Sales).
- Inherited permissions from groups to users.
- Group-based access assignment and reporting.

#### Role Management

- Define roles with granular permissions (e.g., Admin, Editor, Viewer).
- Role hierarchies (e.g., Senior roles inherit from junior ones).
- Time-bound or conditional roles for temporary access.

#### Access Control

- Role-Based Access Control (RBAC) implementation.
- Rule-based permission assignment (based on location, device, etc.).
- Audit trails for access activities.

#### Workflow Automation

- Onboarding/Offboarding workflow with approvals.
- Access request workflow (self-service + manager approval).
- Role change or privilege escalation requests.

#### Workflow Examples

##### User Onboarding Flow

- HR creates a user profile
- System assigns base roles
- Manager approval
- User gets access based on role
- Audit trail created

##### Access Request Flow

- User submits access request
- Manager & Security team approval
- Access granted with expiration

#### Reporting & Dashboards

- Active users by group/role
- Access anomalies and violations
- Role usage frequency
- Workflow approval timelines

#### Documentation & Support

- Admin Guide
- Developer API Reference
- User Training Manual